| EXP.NO: 5 | Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly. |
|---|---|
| DATE:05/03/2025 | |

**AIM:**

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

**ALGORITHM:**

1. **Input:**

- The user fills out a web form with their name and email.
- The form will use both GET and POST methods for data submission.

2. **Steps for Creating and Handling the HTTP GET and POST Requests:**

1. **Create an HTML Form:**
    - Design an HTML form with two input fields: one for the name and one for the email.
    - Create two buttons to allow the user to submit the form using GET or POST methods.
    - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).
2. **Create the Servlet to Handle GET and POST Requests:**
    - Write a Java Servlet class that will handle both GET and POST requests.
    - The servlet will have two methods:
        - doGet() to handle the GET requests.
        - doPost() to handle the POST requests.

- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
   - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
   - Output the data as a response, including the name and email in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
   - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
   - Output the data as a response, displaying the name and email without exposing them in the URL (more secure).
5. **Return the Output:**
   - The Servlet should return an HTML page displaying the name and email entered by the user.
   - For the GET method, the output should be visible in the URL.
   - For the POST method, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
   - Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
   - Test the form using GET and POST methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**

- Test the GET method: The form data is sent in the URL.
- Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

**PROGRAM:**

```
<!DOCTYPE html>
<html>
<head>
<title>Smart Travel Planner</title>

<style>
Body {
   Margin: 0;
   Font-family: “Segoe UI”;
   Background: #eef7ff;
}

Header {
   Background: #0077b6;
   Color: white;
   Padding: 15px;
   Text-align: center;
   Font-size: 24px;
}

.container {
   Width: 85%;
   Margin: 30px auto;
   Display: flex;
   Gap: 20px;
}

.card {
   Flex: 1;
   Background: white;
```

```
    Padding: 20px;
    Border-radius: 10px;
    Box-shadow: 0 0 10px #ccc;
}

H3 {
    Color: #0077b6;
}

Input, select {
    Width: 100%;
    Padding: 10px;
    Margin: 8px 0;
}

Button {
    Width: 100%;
    Padding: 10px;
    Background: #0077b6;
    Color: white;
    Border: none;
}
</style>

</head>

<body>

<header>Smart Travel Planning Website</header>

<div class="container">

<!—GET METHOD →
<div class="card">
<h3>Search Destinations (GET)</h3>

<form action="TravelServlet" method="get">
<input type="text" name="destination" placeholder="Enter
```

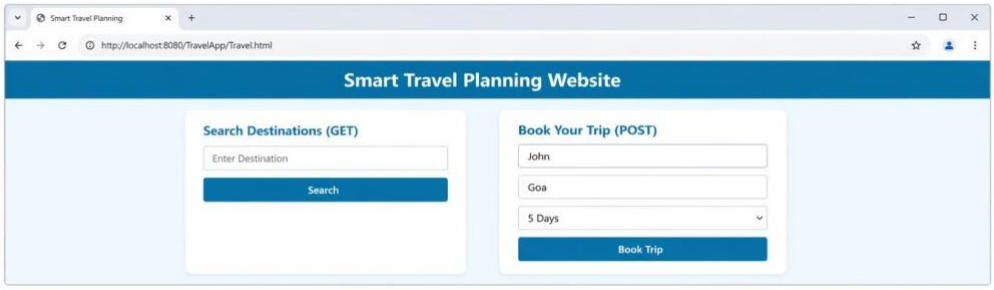
```
Destination”>
<button type=”submit”>Search</button>
</form>

</div>

<!—POST METHOD →
<div class=”card”>
<h3>Book Your Trip (POST)</h3>

<form action=”TravelServlet” method=”post”>
<input type=”text” name=”name” placeholder=”Your Name” required>
<input type=”text” name=”place” placeholder=”Destination” required>

<select name=”days”>
<option value=”3 Days”>3 Days</option>
<option value=”5 Days”>5 Days</option>
<option value=”7 Days”>7 Days</option>
</select>

<button type=”submit”>Book Trip</button>
</form>

</div>

</div>

</body>
</html>
```

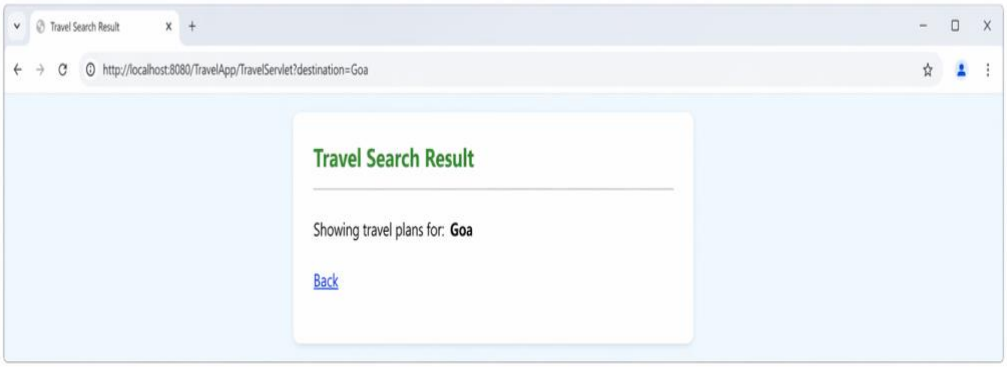
```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;

import javax.servlet.annotation.WebServlet;


@WebServlet("/TravelServlet")
```

```
public class TravelServlet extends HttpServlet {


    protected void doGet(HttpServletRequest request,
HttpServletResponse response)

    throws ServletException, IOException {


        response.setContentType("text/html");

        PrintWriter out = response.getWriter();


        String destination = request.getParameter("destination");


        out.println("<html><body style='font-family:Segoe
UI;background:#eef7ff;'>");

        out.println("<div style='width:500px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");


        out.println("<h2 style='color:green;'>Travel Search Result</h2>");


        if(destination != null && !destination.isEmpty()) {

            out.println("<p>Showing travel plans for: <b>" + destination +
"</b></p>");

        } else {

            out.println("<p>No destination entered</p>");

        }


        out.println("<br><a href='Travel.html'>Back</a>");

        out.println("</div></body></html>");

    }
```

```
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
    throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String place = request.getParameter("place");
        String days = request.getParameter("days");

        out.println("<html><body style='font-family:Segoe
UI;background:#eef7ff;'>");
        out.println("<div style='width:500px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");

        out.println("<h2 style='color:#0077b6;'>Booking Confirmed</h2>");

        out.println("<p><b>Name:</b> " + name + "</p>");
        out.println("<p><b>Destination:</b> " + place + "</p>");
        out.println("<p><b>Duration:</b> " + days + "</p>");

        out.println("<br><a href='Travel.html'>Back</a>");
        out.println("</div></body></html>");
    }
}
```

## Output:


Smart Travel Planning
http://localhost:8080/TravelApp/Travel.html
Smart Travel Planning Website
Search Destinations (GET)
Enter Destination
Search
Book Your Trip (POST)
John
Goa
5 Days
Book Trip


Travel Search Result
http://localhost:8080/TravelApp/TravelServlet?destination=Goa
Travel Search Result
Showing travel plans for: Goa
Back

**RESULT:**

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.